

Daksh K. Shah

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EDUCATION

New York University, Courant Institute

Master of Science in Computer Science (Incoming Fall 2026)

New York, NY

Aug. 2026 – May 2028

University of California, Santa Cruz

B.S. in Computer Science & B.S. Math Theory and Computation — GPA: 3.89

Santa Cruz, CA

Sep. 2022 – June 2026

- **Honors/Awards:** Chancellor's Award for Undergraduate Research, Dean's List (8x)
- **Thesis:** *K-Material SDFs for Neural Attenuation Fields*
- **Selected Coursework:** Computer Graphics, Deep Learning, Numerical Methods, Real Analysis, Advanced Linear Algebra, Database Systems I, Distributed Systems, Algorithm Analysis
- **Graduate Coursework:** Advanced Computer Vision, Statistical Machine Learning, Artificial Intelligence, Advanced ML (NeuroSymbolic AI, Differentiable Programming), Applied Bayesian Statistics

TECHNICAL SKILLS

ML & 3D Computer Vision: 3D Gaussian Splatting, NeRFs, Neural SDFs, Diffusion Models, Vision-Language Models, Self-supervised Learning, LoRA, Hugging Face Transformers

Core Systems & Languages: Python, C/C++, Go, Bash, SQL, Real-Time Systems

Graphics & Compute Libraries: PyTorch, PyTorch Lightning, ONNX, OpenCV, WebGL, OpenGL, Three.js

MLOps & Infrastructure: Docker, Kubernetes, Kubeflow, WandB, Distributed Systems

PUBLICATIONS AND RESEARCH REPORTS

- **Video Super-Resolution Benchmark: Evaluating Spatial Fidelity and Temporal Coherence Tradeoffs** — *Daksh K. Shah, Dylan J. Louie* — Technical Report, UC Santa Cruz (2025)
- **ReNAF: Regularized Neural Attenuation Fields with 3D Reconstruction Priors for Sparse-View CT** — *Emmanouil Nikolakakis, Daksh K. Shah, Julia Wong, Julie Digne, Razvan Marinescu* — In Preparation
- **K-NeAS: Scalable Multi-Material CT Reconstruction using Neural SDFs** — *Daksh K. Shah, Emmanouil Nikolakakis, Razvan Marinescu* — In Preparation for Off-Grid Workshop, MICCAI (2026)

RESEARCH EXPERIENCE

Research Assistant (PI: Razvan Marinescu)

BiomedAI Lab, UC Santa Cruz

January 2024 – Present

Santa Cruz, CA

- Originated **K-Material SDFs** for Neural Attenuation Fields as senior thesis work, later extended into the K-NeAS project (see Projects).
- Supported a PhD student's **ReNAF** publication by designing and running all experiments and authoring the experiment scripts on a **Kubernetes** cluster.

Research Assistant (PI: James Davis)

Visualization and Interactive Systems (VIS) Lab, UC Santa Cruz

April 2025 – June 2026

Santa Cruz, CA

- Began work on an egocentric **Gaussian Splatting** pipeline for Meta Project Aria before transitioning focus to thesis research under Razvan Marinescu.
- Transitioned into an advisory role, mentoring undergraduate researchers and leading the lab's reading group on **Embodied Navigation, Vision-Language Models, and 3D Reconstruction**.

PROJECTS

K-NeAS: Scalable Multi-Material CT Reconstruction | *PyTorch, CUDA, Kubernetes* Nov. 2025 - Present

- Engineered a unified multi-head implicit neural representation with a shared latent backbone and K output heads, achieving up to 1.8 dB improvement in 3D PSNR over baseline NeAS on clinical CBCT data as material count scales from 1 → 4 at 1000 epochs.
- Designed a differentiable K-material soft selector enabling reconstruction across an arbitrary number of tissue types, where baseline methods are hard-capped at 2 materials.
- Improved sparse-view robustness, outperforming baseline by 1.1 dB 3D PSNR at 10-view acquisition settings on clinical Chest CBCT data.
- Automated attenuation bound estimation via unsupervised GMM clustering, eliminating per-scene manual hyperparameter tuning required by the baseline.

MaskSDF: Distilling Physics-Constrained 3D Priors for 2D-to-3D Generation

April 2026 - Present

- Designing a feed-forward **Point-ViT** masked autoencoder with **EG3D-style triplanes**, using a cross-attention bridge to project 3D ViT tokens onto orthogonal 2D feature grids, to reconstruct watertight surfaces from heavily masked ($\geq 80\%$) point clouds without per-scene optimization.

- Enforces physics-grounded geometry via **Eikonal** and surface **normal regularization**, motivated by the hypothesis that a valid SDF triplane latent space is a superior distillation target for 2D-to-3D transfer.
- Cut training cost from 400 to 160 GPU-hours by sharding ~100K files into ~50 **WebDataset** shards and scaling via **DDP** across 4x A6000 GPUs on Kubernetes.

Distributed Sharded Key-Value Store | *Go, Docker*

March 2025

- Engineered a horizontally scalable, sharded key-value storage system in **Go** using custom **RPCs**, implementing **Causal and Eventual Consistency** for distributed synchronization.
- Architected a **fault-tolerant**, partition-resilient cluster deployed via **Docker** for containerized orchestration and automated health monitoring.

PROFESSIONAL EXPERIENCE

Software Engineer Intern

Jan. 2026 – June 2026

Keysight Technologies

Remote

- Migrating legacy WPF plugin interfaces to the KS8500B web app using **C#/.NET** and **Avalonia** to enable cross-platform UI rendering.
- Implementing high-fidelity data visualization to ensure UI parity between desktop and browser-based test automation tools.

LEADERSHIP, TEACHING, & SERVICE

Teaching Assistant & Tutor

April 2023 – June 2026

Baskin School of Engineering, UC Santa Cruz

Santa Cruz, CA

- Supported over 3000 students across 7 academic courses and 10 terms including Graduate Statistical ML, Computer Graphics, Computer Vision, and Deep Learning.
- Engineered automated Piazza mirroring and built student resource sites for **Computer Graphics (CSE 160)**, earning the title “**Best Tutor**” from students for three terms.

Instructional Officer

Aug. 2025 – June 2026

Santa Cruz Artificial Intelligence

Santa Cruz, CA

- Delivered technical workshops on **Neural Network architectures**, training workflows, and practical implementation for 50+ participants.
- Led a mini-hackathon and workshop on MLOps methodologies using kubernetes, docker, and LoRA.